VLSM, Route Summarisation and IP Address Troubleshooting

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OBJECTIVES

By the end of this chapter, you will be able to:

- Understand VLSM and its implementation
- Describe route summarisation
- Understand the basic commands of IP troubleshooting

In the previous chapter, we discussed IP addresses and subnetting. While allocating the IP addresses, the Internet service provider (ISP) faces a challenge. Assume that there is a requirement to allocate IP subnets of different sizes according to the requirement of the customer. For an ISP, it is not feasible to divide the IP addresses into fixed size subnets. But, subnetting the subnets seems like a feasible solution because it results in minimum wastage of IP addresses. The Variable Length Subnet Masks (VLSM) helps in subnetting a subnet thereby reducing the wastage of IP addresses. The VLSM allows route summarisation. In this chapter, you will also learn how to summarise routes to reduce the size of routing tables.

While using IP addresses, many problems can occur due to various reasons, namely, if the client's machine is misconfigured, malfunctioning, or in extreme cases, shuts down. The IP troubleshooting section discusses all the issues related to the IP addresses and how to troubleshoot them.

Variable Length Subnet Masks

The VLSM, which is defined in RFC 1812, helps in subnetting a subnet and allows you to use different subnet masks for different networks. By using the VLSM, wastage of IP addresses get reduced.

CERTIFICATION OBJECTIVE

Understand VLSM and its implementation

The variable length subnetting creates variable length subnet masks and deploys different size subnets for a given network address. For example, 255.255.255.252 is a good mask for a point-to-point link because it provides two hosts for each subnet. 255.255.255.192 is a good mask for a LAN connection because it provides 62 host addresses. Using 255.255.255.252 mask for a LAN connection will not be sufficient and using 255.255.255.192 for point-to-point connection will waste the addresses. The VLSM solves this problem by using different subnet mask values in the same class address space.

The VLSM has two main advantages:

- It helps to use the address space efficiently in large scale networks.
- It helps to reduce the size of routing tables which is known as route summarisation.

Figure 1 shows you a network which uses subnetting to allot the IP addresses. In Figure 1, the Router A has point-to-point connections with Router B, Router C, and Router D. Let us suppose that the LAN segments at these routers have 50 devices. So, a subnet mask 255.255.255.192/26 can be used. This mask provides 62 hosts per subnet. You can allot only four blocks of subnets 0, 64, 128, and 192. Therefore, a another Class C network (192.168.2.0/26) is required. This allotment wastes a lot of addressing space at the point-to-point connection. The subnets 192.168.1.128 and 192.168.1.196 are wasted as they are not being used.

IMPORTANT NOTES

Only classless protocols, such as EIGRP, OSPF, and RIPv2 support the VLSM with route summarisation.

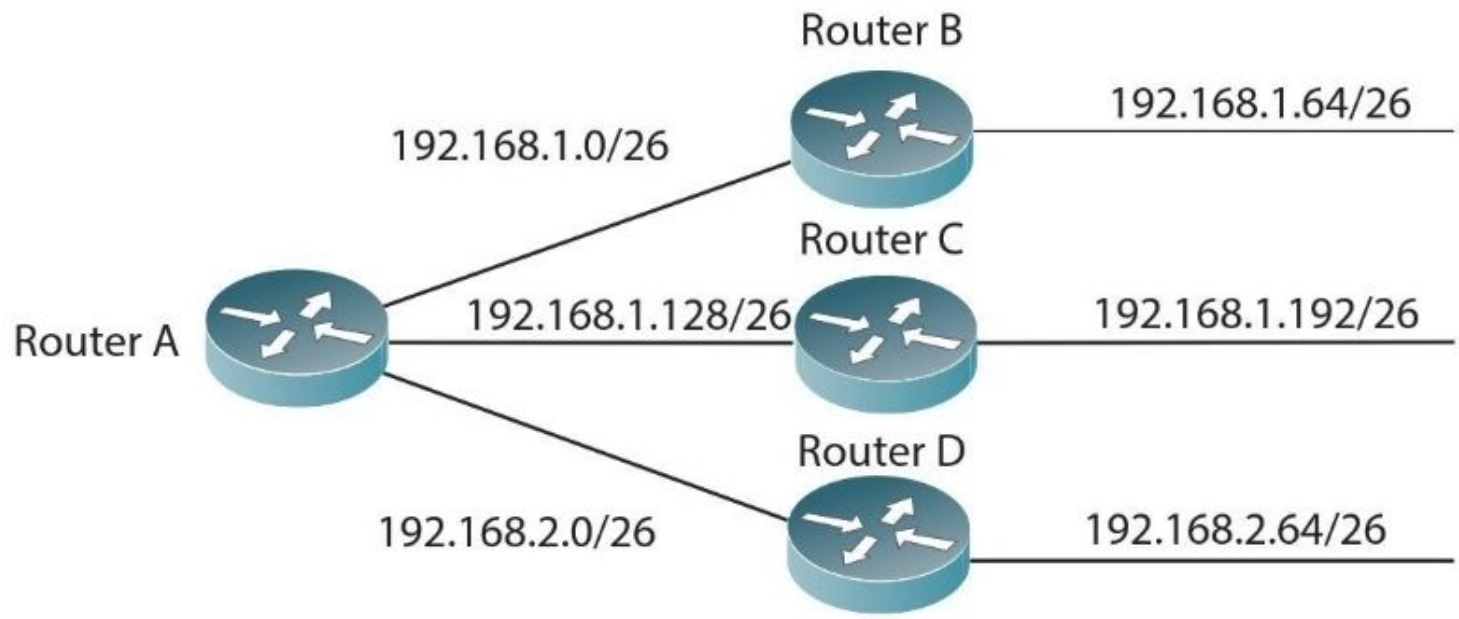


Figure 1: Before VLSM

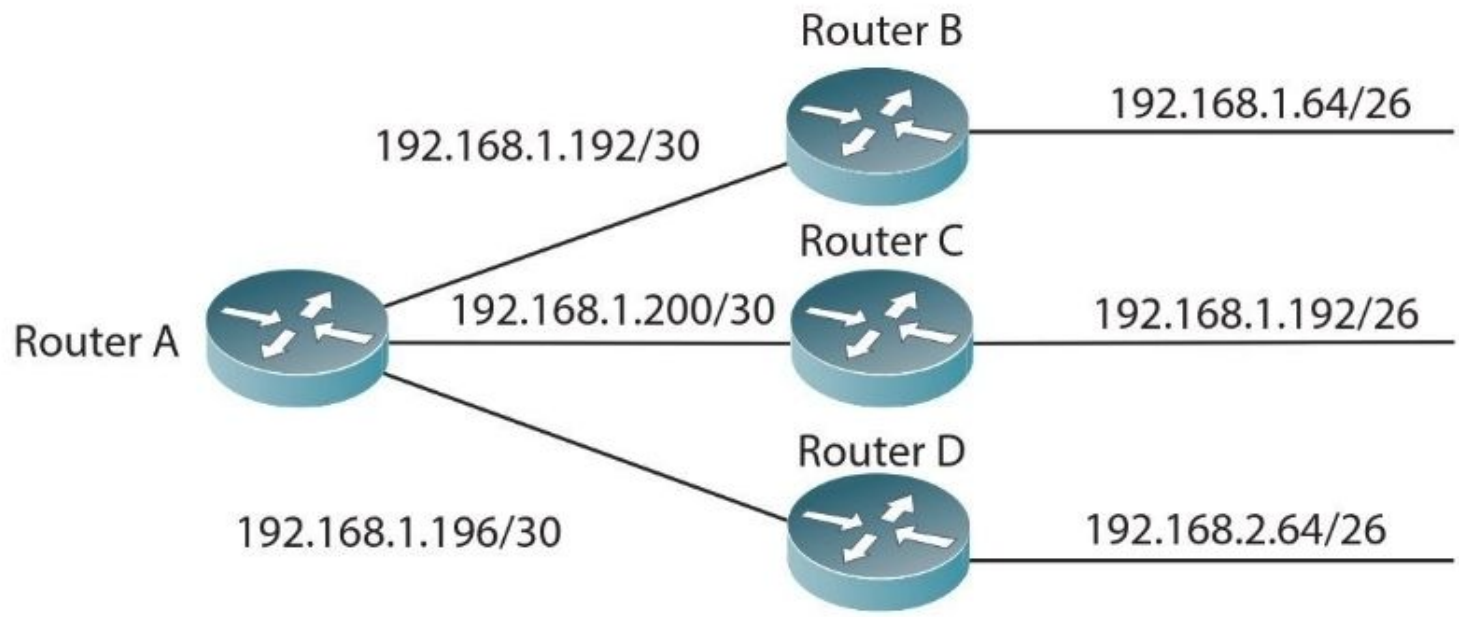


Figure 2: After VLSM

Now, Figure 2 shows an efficient usage of address space using the VLSM. In this figure, Router B, Router C, and Router D use a 192.168.1.192/26 subnet, but for the point-to-point connections between Router A to Router B, Router C, and Router D, a different subnet is used. This helps in using the address space efficiently.

VLSM Networks Implementation

To implement the VLSM, complete the following steps:

1. Identify the largest network with the maximum number of hosts connected to it.

2. Find an appropriate subnet mask for that network.
3. Write the appropriate subnet blocks to fit the subnet mask.
4. For smaller networks, take a newly-created subnet mask and apply a different and appropriate subnet mask to it.
5. Write all the new subnetted subnets.
6. For much smaller networks, repeat steps starting from step 4.

Let us understand the VLSM implementation using an example. Suppose, you have three LAN segments: one with 74 hosts, second with 60 hosts, and the third with 27 hosts.

Step 1 and 2: The largest segment has 74 hosts. 192.168.1.0/25 subnet mask gives you two subnets of 126 hosts.

Subnet	Subnet 1	Subnet 2
Network ID	192.168.1.0	192.168.1.128
First Host Address	192.168.1.1	192.168.1.129
Last Host Address	192.168.1.126	192.168.1.254
Broadcast ID	192.168.1.127	192.168.1.255

Step 3: The newly-created subnets are: 192.168.1.0/25 and 192.168.1.128/25. Assign the first subnet 192.168.1.0/25 to the LAN segment having 74 hosts.

Step 4: The next LAN segment has 60 hosts. First, the subnet is assigned. Now, the second segment is available. To save addresses, we will subnet the second subnets. /26 provides us 4 subnets each having 62 host addresses.

Step 5: The newly created subnets are: 192.168.1.0/26, 192.168.1.64/26, 192.168.1.128/26, and 192.168.1.192/26. Assign 192.168.1.128/26 to the LAN segment having 60 hosts. We cannot assign 192.168.1.0/26 and 192.168.1.64 because they are already assigned to the largest LAN.

Step 6: Now, we are left with our last segment having 27 hosts and the mask has 62 hosts, which is not the most efficient one. You can go back to step 4 and repeat the process for this subnet. The subnet mask /27 (255.255.255.224) has 30 host addresses. We will assign the subnet 192.168.1.192/27 to it.

VLSM Example 1

Figure 3 illustrates a much more difficult network than the above example shown in Figure 2. In this example, a Class C network is provided. Your task is to allot the IP addresses using the VLSM.

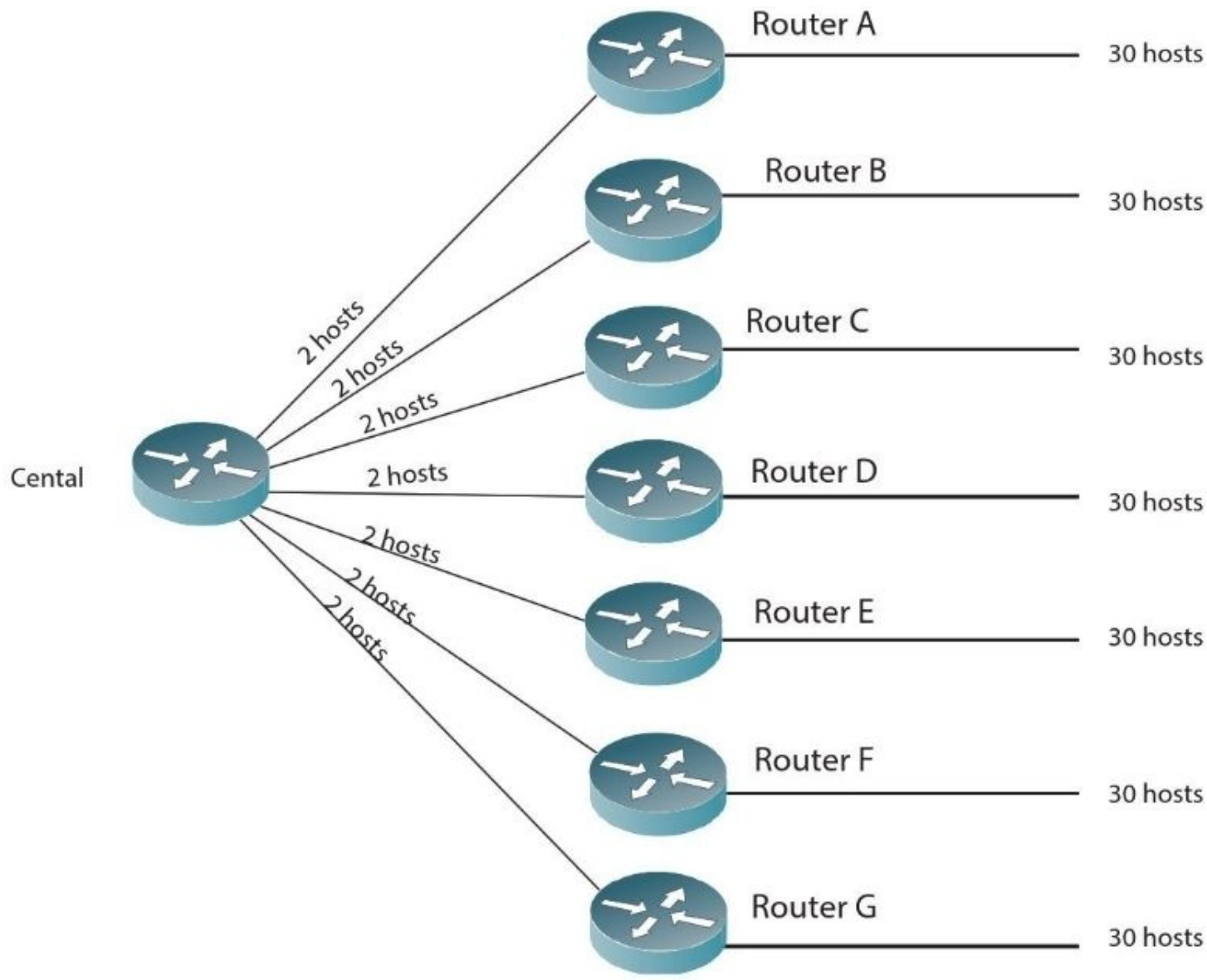


Figure 3: Sample network 1

First, you will assign the address space to the segments having the largest number of nodes. In Figure 3, there are seven segments that have 30 hosts. To allot an IP address, we will create a subnet mask of 255.255.255.224 (/27). It will create eight subnets: 192.168.2.0/27, 192.168.2.32/27, 192.168.2.64/27, 192.168.2.96/27, 192.168.2.128/27, 192.168.2.160/27, 192.168.2.192/27, and 192.168.2.224/27. Assign first seven subnets to them and use the last subnet 192.168.2.224/27 to create more subnets for point-to-point links. We will create a subnet mask of 255.255.255.254 (/30). This will create 64 small subnets of 2 hosts each. Assign

the first seven subnets to the point-to-point links. Figure 4 shows the layout of a sample network 1 using the VLSM.

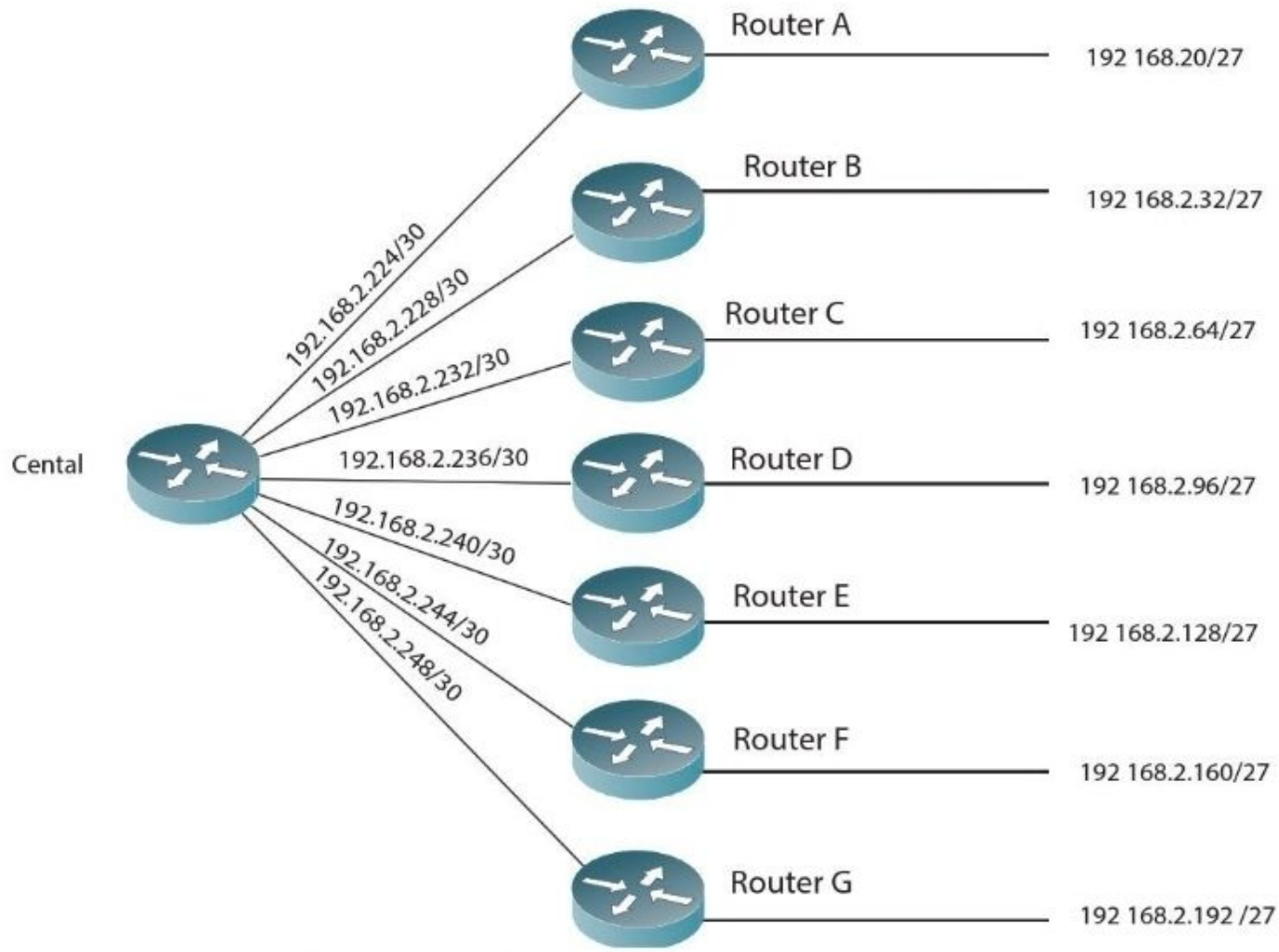


Figure 4: Sample Network 1 using VLSM

VLSM Example 2

Figure 5 demonstrates another sample network. In this example, a Class C network is provided. Your task is to allot the IP addresses using the VLSM.

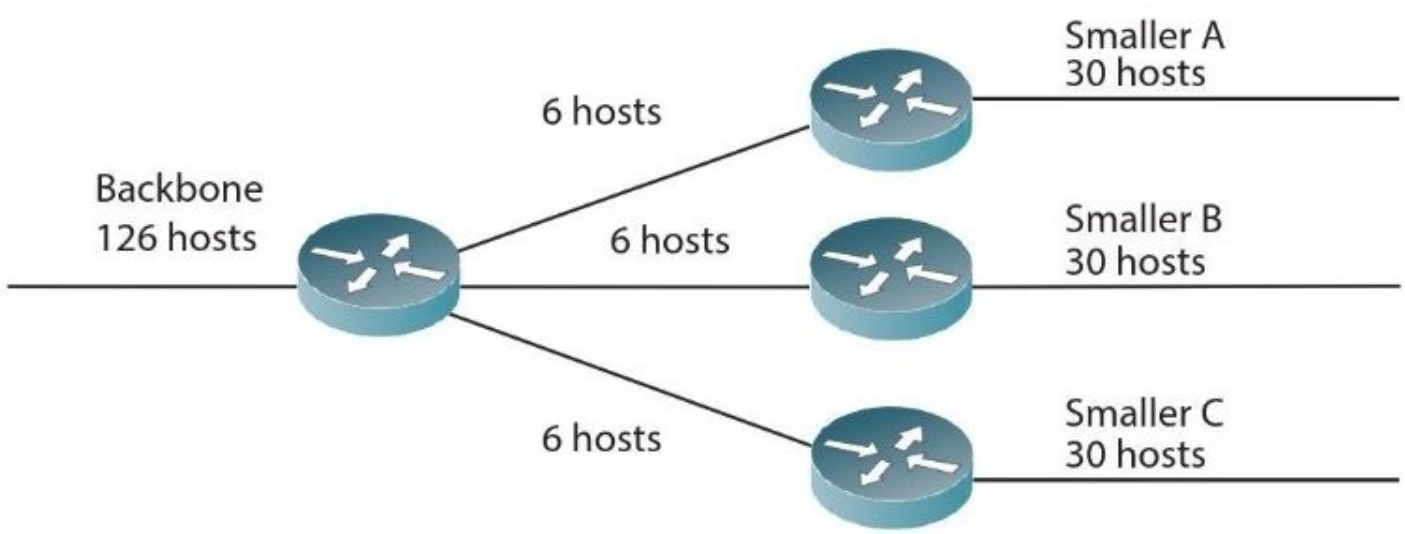


Figure 5: Sample Network 2

First, you will assign the address space to the segments having the largest number of nodes. In Figure 5, the largest network segment is the backbone network having 126 hosts. To allot an IP address, we will create a subnet mask of 255.255.255.128 (/25), which will create two subnets: 192.168.2.0/25, and 192.168.2.128/25. Assign the first subnet to it.

Then, we will use the second subnet 192.168.2.128/25 to create more subnets. We will create a subnet mask of 255.255.255.224 (/27). This will create subnets: 192.168.2.128/27, 192.168.2.160/27, 192.168.2.192/27, and 192.168.2.224/27. Assign the first three subnets to the LAN segments. Use the remaining subnet for point-to-point connections. Figure 6 illustrates the layout of sample network 2 using the VLSM.

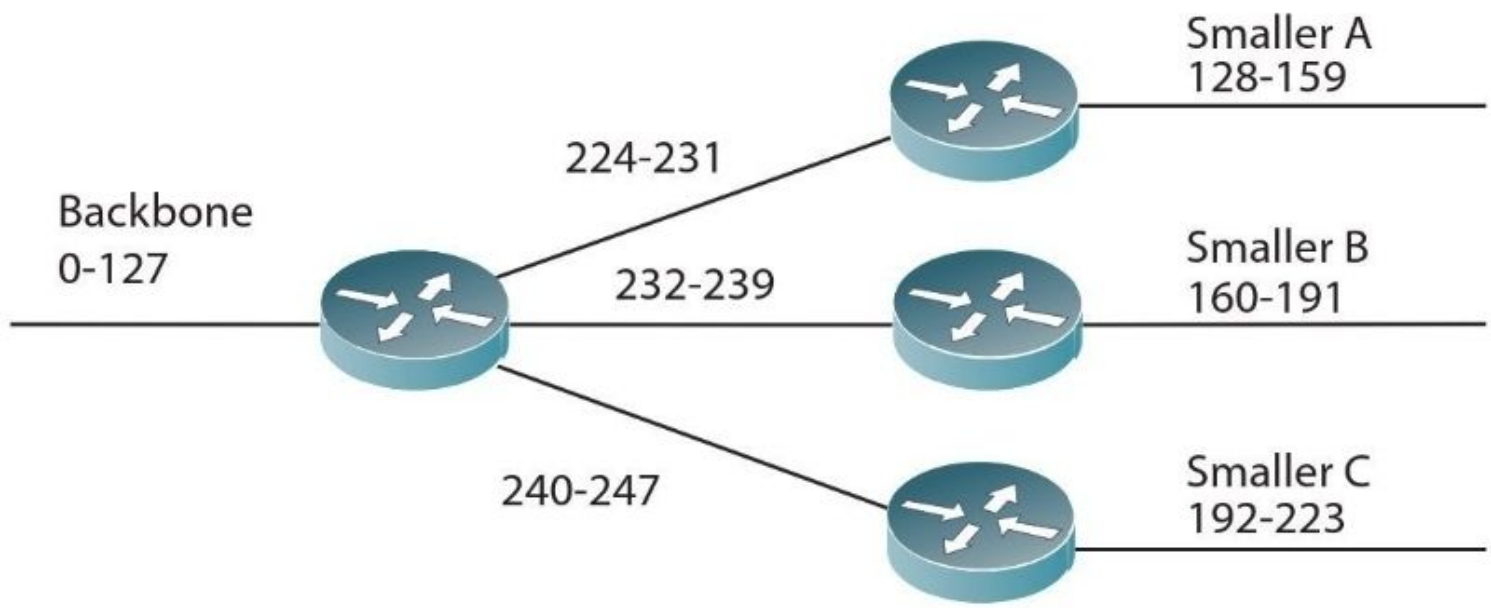


Figure 6: Sample Network 2 using VLSM

Route Summarisation

The VLSM performs route summarisation thereby allowing routing protocols to take many networks and advertise them as one address. It is also called as route aggregation. It reduces the size of routing tables on routers to save memory. It also reduces the amount of time taken by the IP to parse the routing table and find the path. It has the following advantages:

CERTIFICATION OBJECTIVE

Describe route summarisation

- Size of routing tables is reduced, which requires less memory and processing.
- Size of routing updates is reduced, which requires less bandwidth

Figure 7 illustrates the route summarisation. It shows that all the IP addresses are summarised into 10.0.0.0/8.

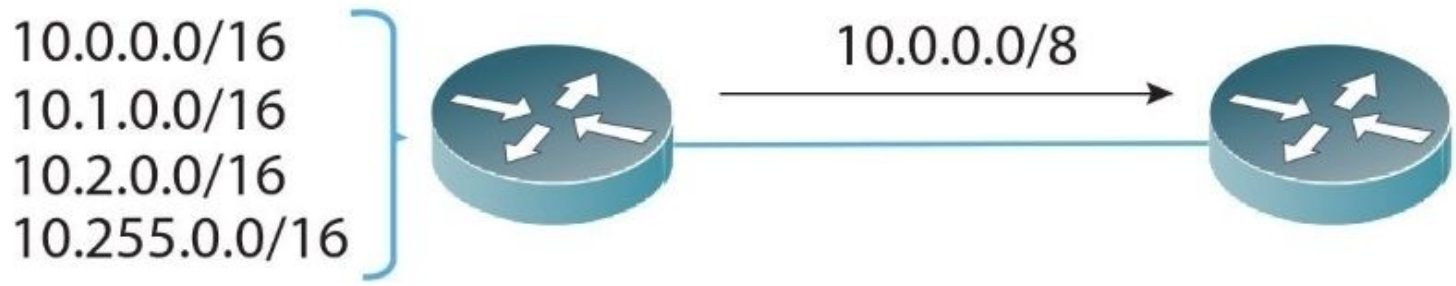


Figure 7: Route Summarisation

Let us take another example. Suppose, you want to summarise the following networks as one network advertisement.

192.168.16.0 through network 192.168.31.0

First, you need to find the block size. There are exactly 16 Class C networks, so this fits into a block size of 16. Now, with the help of the block size, you can find the network address and mask used to summarise these networks into one advertisement. The first network address in the block is always used as a summary address. In this example, the first summary address is 192.168.16.0.

The next step is to find out the block size. From the subnet mask 240, a block size of 16 can be obtained. 240 comes in the third octet. So, the mask would be 255.255.240.0.

Let us take an example from a host's view.

A summary address is 192.168.144.0/20. What is the range of host addresses that would be forwarded according to this summary?

The /20 provides a summary address of 192.168.144.0 and mask of 255.255.240.0. The third octet has a block size of 16 and starts at the summary address of 144. The next block of size 16 is 160, so the network summary range is 144 to 159 in the third octet. A router will forward any packet with destination IP addresses of 192.168.144.1 through 192.168.159.254.

ANSWER IT

1. What is the other term for route aggregation?
   a. VLSM
   b. Load balancing
   c. Subnetting
   d. Summarisation
2. Which mask should be used to connect a point-to-point link to reduce wastage of IP addresses?
   a. /27
   b. /28
   c. /29
   d. /30

IP Address Troubleshooting

Let us suppose, you are a host and you are not able to connect to a remote server. What will you do? Troubleshooting IP addressing helps in solving these kinds of problems. Figure 8 shows you a sample network for IP troubleshooting.

CERTIFICATION OBJECTIVE

Understand the basic commands of IP troubleshooting

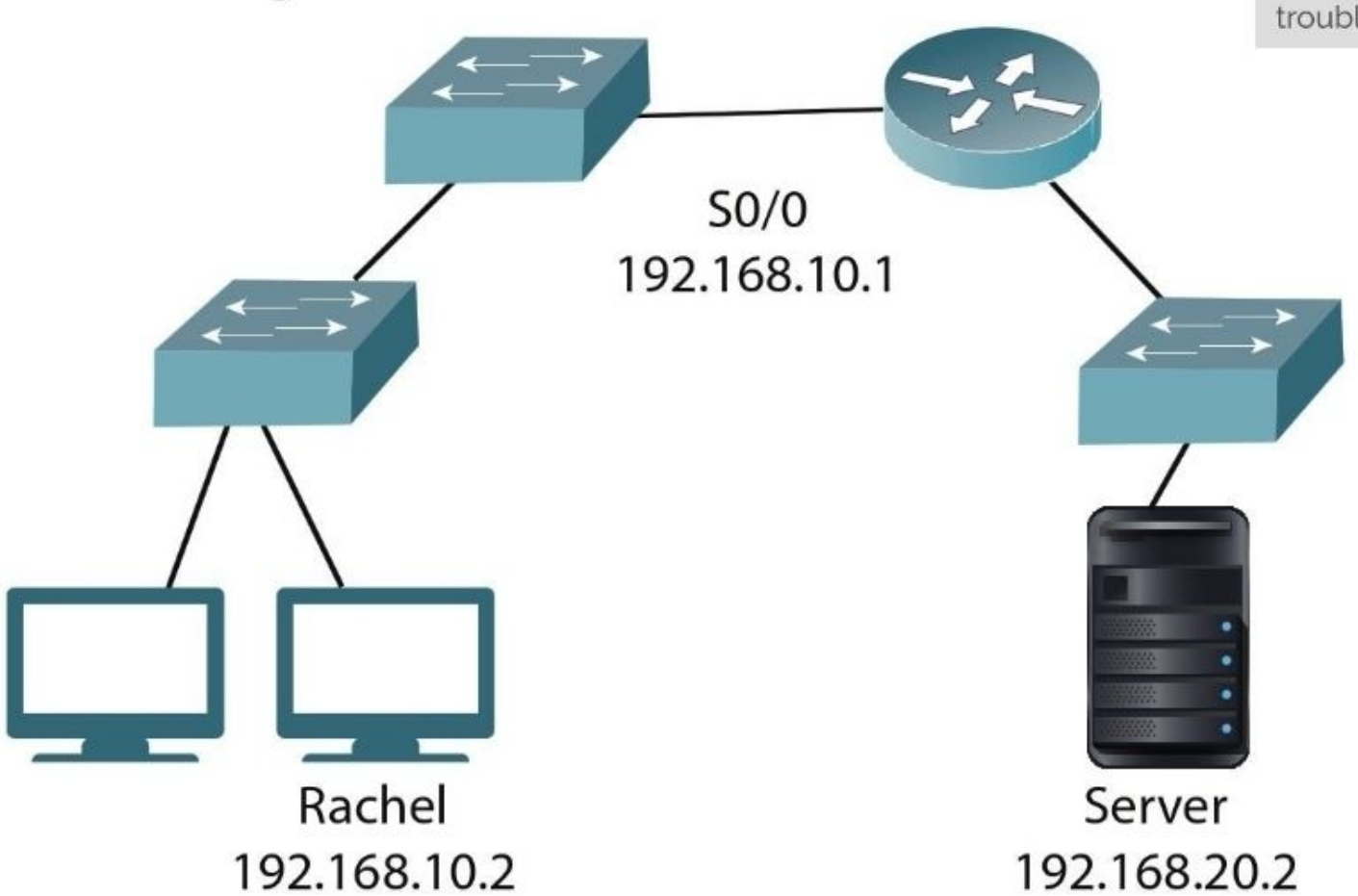


Figure 8: IP troubleshooting

Cisco follows the following troubleshooting steps:

1. Open the command window and ping 127.0.0.1. This is a loopback address. If you get a successful ping, your IP stack has no problem. If it fails, then you need to reinstall the TCP/IP on the host. Figure 9 shows the ping to the loopback address.

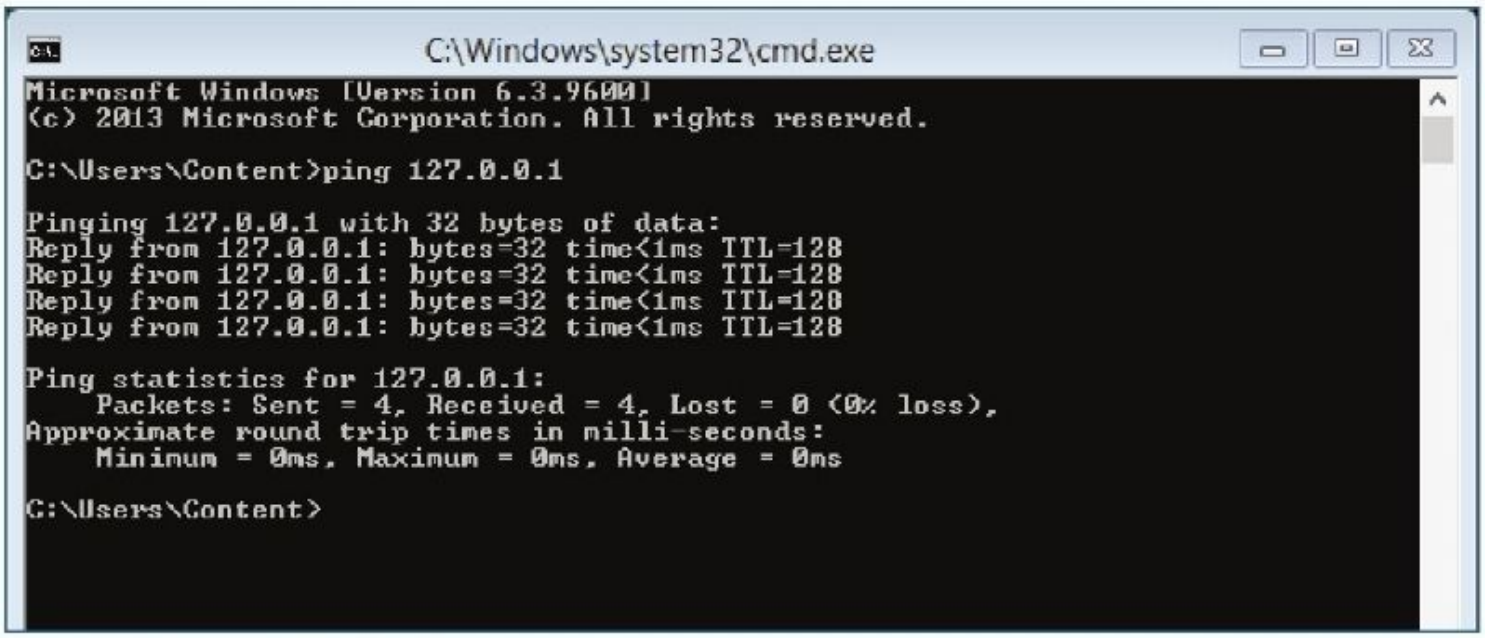


Figure 9: Ping to loopback address

2. Open the Command window and ping the IP address of the local host. If you get a successful ping, your NIC has no problem. If it fails, then there is a problem with the NIC. Figure 10 shows the ping to the local host.

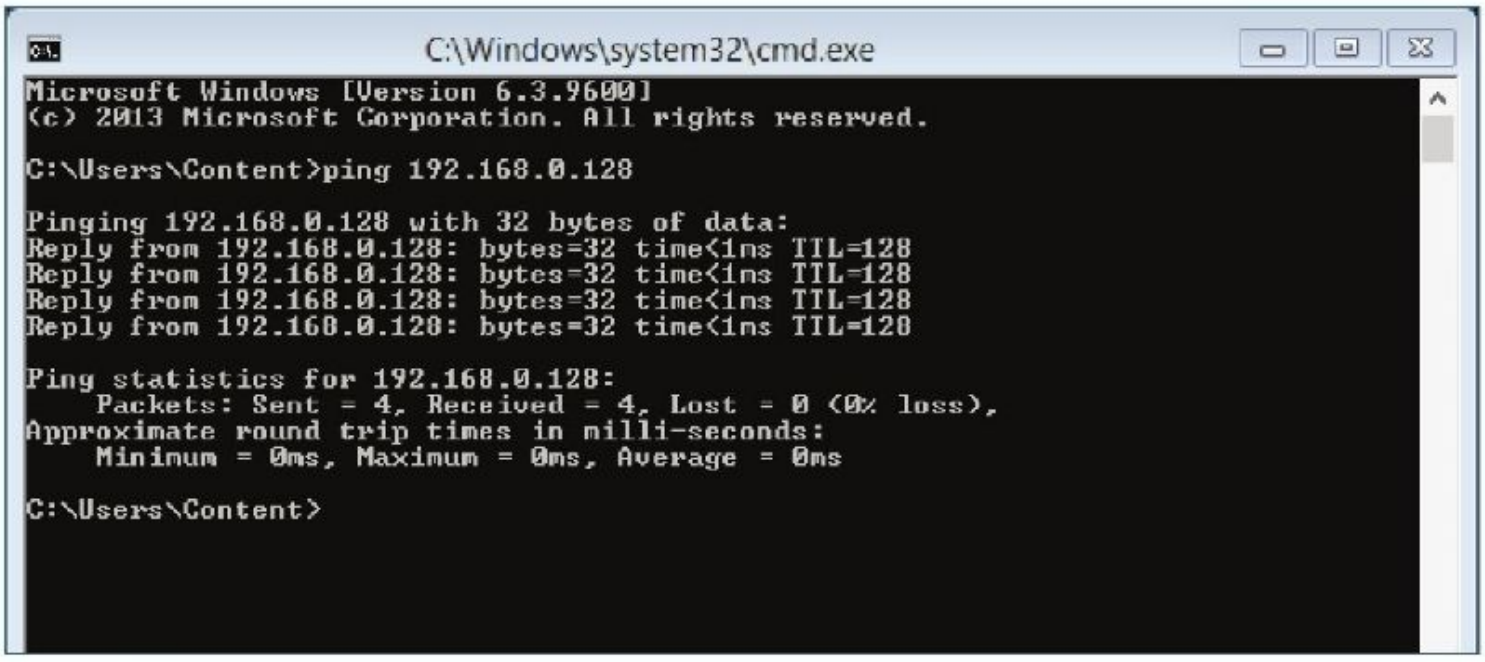


Figure 10: Ping to local host

3. Open a Command window and ping the IP address of the default gateway. If you get a successful ping, your NIC can communicate to the local network. If it fails, then there is a problem with the local network between the NIC and the router. Figure 11 shows the ping to the default gateway.

```
C:\Windows\system32\cmd.exe
Microsoft Windows [Version 6.3.9600]
(c) 2013 Microsoft Corporation. All rights reserved.

C:\Users\Content>ping 192.168.0.1

Pinging 192.168.0.1 with 32 bytes of data:
Reply from 192.168.0.1: bytes=32 time<1ms TTL=64
Reply from 192.168.0.1: bytes=32 time=1ms TTL=64
Reply from 192.168.0.1: bytes=32 time=1ms TTL=64
Reply from 192.168.0.1: bytes=32 time=1ms TTL=64

Ping statistics for 192.168.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\Users\Content>
```

Figure 11: Ping to default gateway

4. Open a Command window and ping the IP address of the remote. If you get a successful ping, then there is a communication between the local host and the remote server. Figure 12 shows the ping to the remote server.

```
C:\Windows\system32\cmd.exe
Microsoft Windows [Version 6.3.9600]
(c) 2013 Microsoft Corporation. All rights reserved.

C:\Users\Content>ping 192.168.0.130

Pinging 192.168.0.130 with 32 bytes of data:
Reply from 192.168.0.130: bytes=32 time=20ms TTL=128
Reply from 192.168.0.130: bytes=32 time<1ms TTL=128
Reply from 192.168.0.130: bytes=32 time=1ms TTL=128
Reply from 192.168.0.130: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.0.130:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 20ms, Average = 5ms

C:\Users\Content>
```

Figure 12: Ping to remote server

Some of the basic commands that help in troubleshooting a network are:

- **`ping`**: ping stands for Packet InterNet Groper. It uses an ICMP echo request and replies to test if the connection is alive or not.
- **`traceroute`**: It displays the list of routers on a path to a network destination. This command does not work from a command prompt.
- **`tracert`**: It works in the same way as a traceroute. But, it is a Microsoft Windows command and will not work on a Cisco router.
- **`arp -a`**: It displays the locally resolved IP addresses as MAC addresses.
- **`show ip arp`**: It works similar to arp -a, but displays the ARP table on a Cisco router.
- **`ipconfig /all`**: It is the host's network configuration.

Recap

- VLSM enables you to create different subnets of specific sizes and it allows you to subnet the subnets.
- VLSM is supported by classless protocols, such as Border Gateway Protocol (BGP), Open Shortest Path First (OSPF), and Routing Internet Protocol version 2 (RIPv2).
- VLSM allows route summarisation, which advertises many networks as one.
- Route summarisation has advantages like reducing the routing table size and reducing the bandwidth required for routing updates.
- The four simple steps to troubleshoot the IP connections are: ping the loopback address, ping the local address, ping the default gateway address, and ping the remote server address.

Test Your Skills

Multiple Choice Questions

1. Which mask should be used on a point-to-point links to reduce the waste of IP address using VLSM? /27
   a. /28
   b. /29
   c. /30
2. Which two subnetworks would be included in the summarised address of 172.31.80.0/20? (Choose two)
   a. 172.31.17.4/30
   b. 172.31.80.0/22
   c. 172.31.92.0/22
   d. 172.31.192.0/18
3. Why is summarisation so important to an efficient routed system?
   a. It adds detail to the route tables of routers.
   b. Summarisation sends all subnets as classful networks, eliminating the overhead of transmitting the mask in routing updates.

c. Summarisation reduces the size of the route tables, prevents route table instability due to flapping routes, and reduces the size of routing updates.

d. Summarisation enforces router authentication, preventing spurious updates from excessively loading the router.

4. Which of the following devices share the same network?
   a. 192.168.78.25/29
   b. 192.168.78.23/29
   c. 192.168.78.33/29
   d. 192.168.78.38/29

5. Which of the following would best support a point-to-point link?
   a. 255.255.255.253
   b. 255.255.255.0
   c. 255.255.255.128
   d. 255.255.255.252

6. Of the following, which network address is always used to advertise a summary address?
   a. The last network address in the block
   b. The next to last network in the block
   c. The second network in the block
   d. The first network in the block

7. When a ping to the loopback address fails, what can you assume?
   a. The IP address of the local host is incorrect.
   b. The IP address of the remote host is incorrect.
   c. The NIC is not functional.
   d. The IP stack has failed to initialise.

8. Which protocol is use by the ping command?
   a. DHCP
   b. ICMP

c. ARP

d. DNS

9. Which of the following commands displays the networks traversed on a path to a network destination?

   a. `Ping`

   b. `Traceroute`

   c. `Pingroute`

   d. `Pathroute`

10. Which command generated the output shown below?

```
Reply from 192.168.20.1: bytes=32 time<1ms TTL=128
Reply from 192.168.20.1: bytes=32 time<1ms TTL=128
Reply from 192.168.20.1: bytes=32 time<1ms TTL=128
Reply from 192.168.20.1: bytes=32 time<1ms TTL=128
```

   a. `Traceroute`

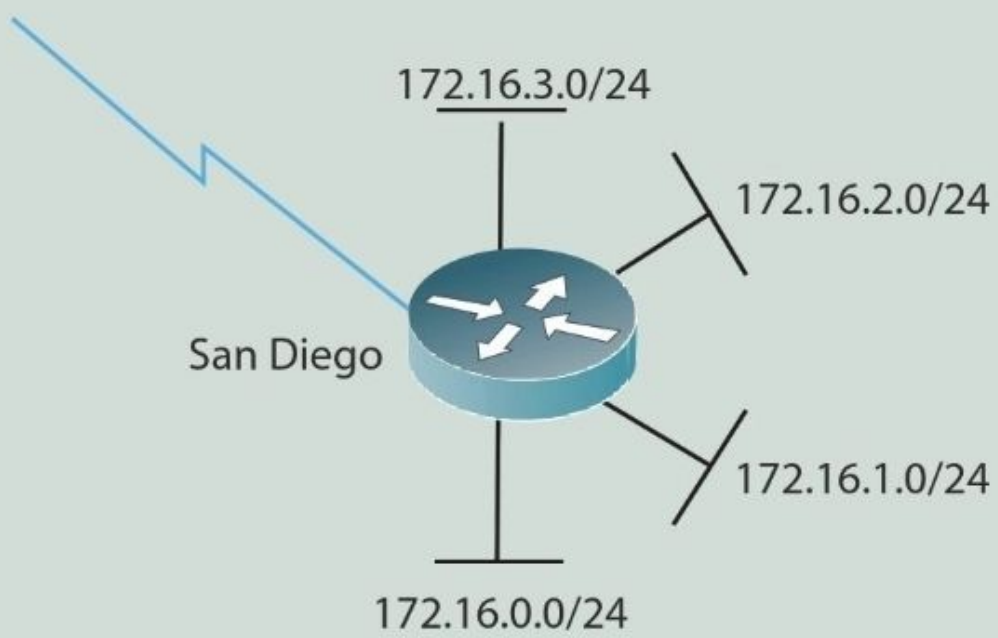
   b. `show ip route`

   c. `ping`

   d. `pathping`

Scenario Based Questions

1. The administrator wants to reduce the size of the central router routing table. Which of the following summary routes would represent all the LANs in Tempe, with no additional subnets?



a. 172.16.0.0/24
b. 172.16.0.0/23
c. 172.16.0.0/22
d. 172.16.1.0/24

2. An Ethernet port on a router in your organisation is assigned the IP address 10.65.64.1/21. What is the maximum number of hosts allowed on this subnet?
   a. 254
   b. 510
   c. 1022
   d. 2046
   e. 4094
   f. 4096

3. What single summary address could be used for the following networks?
   a. 192.168.32.0/19
   b. 192.168.0.0 255.255.240.0
   c. 192.168.64.0/18

4. What summary address would cover all the networks shown and advertise a single, efficient route to Router B that won't advertise more networks than needed?
   a. 172.16.0.0/24
   b. 172.16.1.0/24
   c. 172.16.0.0/24
   d. 172.16.0.0/20